

Chapter 4

Buckling of Columns

4.1 Question Bank UNIT - 1 AS

Quiz Questions

1. Increasing the slenderness ratio the critical buckling load.
2. Rankine-Gordon's empirical formula is applicable for
3. Euler's formula is applicable only
4. Slenderness ratio is the ratio of effective length of column and
5. What is the relation between actual length and effective length while determining crippling load for a hollow rectangular cast iron column having both ends fixed?
6. While determining crippling load, the effective length of solid circular bar is $\frac{1}{\sqrt{2}}$ of actual length if,
7. What is the safe load acting on a long column of 2 m having diameter of 40 mm. The column is fixed at both the ends and modulus of elasticity is 200 GPa? (F.O.S = 2)
8. If the effective length of a column is twice the actual length, then the column is
9. With identical beam and column, buckling occurs as compared to bending under a load
10. Nature of stresses produced in buckling and bending are the
11. Keeping loading same but increasing the length, normal stresses in a beam will
12. Keeping loading same but increasing the length, shear stresses in a beam will
13. A column of length 4m with both ends fixed may be considered as equivalent to a column of lengthwith both ends hinged.
14. Rankine's formula is generally used when slenderness ratio lies in between
15. Euler's formula is not valid for mild steel column when slenderness ratio is
16. An electric pole is 6.5 m high from the ground level. Its effective length for design purposes will be

17. Calculate the Euler's crippling load, if the effective length of column is 10 m take flexural rigidity as 6.14×10^{10} N/mm²
18. Columns with what slenderness ratio are not designed with respect to buckling but are designed for compressive stresses.
19. If slenderness ratio=45, which mode of failure will dominate?
20. Cast iron column with a slenderness ratio of 75 are columns
21. Columns with what slenderness ratio are not designed with respect to buckling but are designed for compressive stresses.

Main Questions

1. An aluminum ($E = 70$ GPa) column built into the ground has length, $L = 2.2$ m, and is under axial compressive load P . The dimensions of the cross-section are $b = 210$ mm and $d = 280$ mm. Calculate The critical load to buckle the column.
2. Differentiate between elastic buckling and Inelastic buckling of columns.
3. What is the safe load acting on a long column of 2 m having diameter of 40 mm. The column is fixed at both the ends and modulus of elasticity is 210 Gpa? (F.O.S = 2) Describe the loading on the beam.
4. Explain Euler's theory of columns stability, write assumptions and limitations.
5. The buckling load for a column hinged at both ends is 10 kN. If the ends are fixed, the buckling load changes to what value?
6. What is the relation between actual length and effective length while determining crippling load for a hollow rectangular cast iron column having both ends fixed?
7. Derive an expression for buckling equation for a pin-supported long slender column.
8. Describe critical load and stability of columns.
9. Determine the ratio of buckling strengths of two columns one hollow and the other solid. Both are made of the same material and have the same strength cross sectional area and end conditions. The internal diameter of hollow column is half of its external diameter.
10. A hollow cylindrical cast iron column is 4m long with both ends fixed. Determine the minimum diameter of the column if it has to carry a safe load of 250kN with a factor of safety of 5. Take the internal diameter as 0.8 times the external diameter.
 $\sigma_c = 550$ MPa and $a = 1/1600$ in rankines formula.

11. Design a circular column by specifying the diameter and length, of a steel column ($E=30 \times 10^6 \text{ psi}$) that will carry a 50 kip load with a safety factor between 2 and 10. The design requirements are that it is in stable equilibrium and that bending will not be the failure mode. Assume the ends are pinned at both ends
12. Find the best material based on Young's Modulus that prevents the column from buckling for the following condition's
Length: 15 feet
Diameter: 3 inches
Load: 5000 lb
FS: 2.0
End effects: Pinned - Pinned
13. A round steel rod of diameter 15 mm and length 2m is subjected to a gradual increasing axial compressive load. Using Euler's formula find the buckling load. Find also the maximum lateral deflection corresponding to the buckling condition. Both ends of the rod may be taken as hinged. Take $E = 210 \text{ GPa}$ and the yield stress of steel = 240 MPa.
14. A cast iron hollow cylindrical column 3 m in length when hinged at both ends, has a critical buckling load of $P \text{ kN}$. When this column is fixed at both the ends, its critical load rises to $P+300 \text{ kN}$. If the ratio of external diameter to internal diameter is 1.25 and $E = 100 \text{ GPa}$. Determine the external diameter of the column.